

CHEN-YE HONG

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EDUCATION

National Taiwan University (NTU)

Sep. 2022 – Dec. 2025 (Expected)

B.S. in Mechanical Engineering

Taipei, Taiwan

- Overall GPA: 4.16/4.3; Rank: 5/210 (Top 2%)
- Relevant Courses: Automatic Control (A+), Applied Electronics (A+), Digital Control System (A+)
- NTU Leadership Program 13th

National Tsing Hua University (NTHU)

Sep. 2021 – Jun. 2022 (Transferred to NTU)

B.S. in Power Mechanical Engineering

Hsinchu, Taiwan

- Overall GPA: 4.16/4.3; Rank: 1/116

RESEARCH INTERESTS

Human-Robot Interaction (HRI), Assistive Robotics, Human Intent Recognition and Prediction

PUBLICATIONS

- **Chen-Ye Hong**, Chih-Hao Chou, Yi-Ji Wang, Kuan-Lin Wu, Chih-Hao Chang, and Dian-Ru Li (2025). “Development of a Smart Cadaveric Limb Motion Simulator for Physiologically Relevant In-Vitro Biomechanical Evaluation of Small Joints.” *Submitted to IEEE Transactions on Biomedical Engineering*. (Under Review; First co-author)
- Kuan-Lin Wu, **Chen-Ye Hong**, Chih-Hao Chou, Yi-Ji Wang, and Dian-Ru Li (2025). “Smart Joint Motion Simulation Platform with Integrated Force Sensing for Clinical Assessment of Artificial Finger Joints.” *Proceedings of the 42nd National Conference on Mechanical Engineering of CSME*. (First co-author, Oral Presenter)
- Chiao-Yu Hsieh, **Chen-Ye Hong**, Jia-Zhen Li, Pei-Chieh Yang, and Yu-Te Wang (2025). “From Movement to Manipulation: Multimodal Control of a Mobile Assistive Robot.” *Proceedings of the 30th International Conference on Technologies and Applications of Artificial Intelligence (TAAI 2025)*. (Accepted for Extended Abstract; First co-author)

RESEARCH EXPERIENCE

Human-Centric Autonomous Driving: VR-Based Internal State Modeling

Sep. 2025 – Present

Research Assistant, BCI Lab (Supervisor: Dr. Yu-Te Wang), Academia Sinica (CITI)

- Designed a Perceptual Modulation Framework to buffer passenger anxiety in high-dynamic scenarios, laying the groundwork for internal-state-aware HRI systems.
- Constructing a multisensory VR testbed to manipulate driving dynamics and environments, developing a computational model of passenger internal states for future control integration.

Multimodal Control for an Assistive Mobile Manipulator

Jul. 2025 – Aug. 2025

Summer Intern, BCI Lab (Supervisor: Dr. Yu-Te Wang), Academia Sinica (CITI)

- Architected a novel multimodal control system that fuses BCI for navigation with gesture recognition for 6-DOF manipulation, resolving modal contention via an EMG-based switch.
- Completed a time-constrained (<15 min) multiple-object retrieval task, achieving 84% BCI decoding accuracy.

Data-Driven Kinematic Stability Assessment of Artificial Joints

Jan. 2024 – Oct. 2025

Undergraduate Researcher, SAM Lab (Supervisor: Prof. Dian-Ru Li), NTU

- Established a standardized, data-driven protocol to quantify artificial joint kinematic stability from clinical video.
- Formulated the core stability metrics by applying k-means clustering and strain field analysis to identify and quantify high-strain regions purely from non-contact motion data.

Physics-Based VR Simulator for Telerobotic Surgical Training

Jul. 2023 – Aug. 2024

Undergraduate Researcher, Advanced Medical Device Lab (Supervisor: Prof. Hao-Ming Hsiao), NTU

- Designed a VR simulation environment in Unity as an intuitive training platform for telerobotic arterial surgeries.
- Modeled physics-based surgical interactions, including deformable arterial plaque, stent deployment via balloon inflation, and guide wire dynamics, and deployed the simulator on the Meta Quest 2.

ACADEMIC PROJECT

Low-Cost 3D Perception and Navigation for Autonomous Vehicles

Feb. 2025 – Jun. 2025

Design and Practice of Intelligent Vehicles at NTU

- Achieved 75% accuracy for 3D object detection by engineering a low-cost perception pipeline integrating a ZED stereo camera with a fine-tuned YOLOv8.
- Implemented a geometry-based path planning algorithm that computes a target trajectory from the 3D locations of detected objects, enabling basic autonomous navigation.

INDUSTRY EXPERIENCE

Hewlett-Packard Company (HP)

Jul. 2024 – Jun. 2025

Mechanical Engineering Intern in PC Design Team

Taipei, Taiwan

- Engineered a novel laptop hinge to improve system cooling, achieving up to a 1°C reduction in surface temperature.
- Reduced replacement time by 86% and improved repairability by designing a modular SSD shielding cap for use across three product lines.

LEADERSHIP & COMMUNITY ENGAGEMENT

National Taiwan University Mechanical Engineering International Compass (NTUMEIC)

Sep. 2024 – Present

Founder & President

- Founded a nonprofit organization to bridge the information and economic gap for NTU students pursuing global academic and career opportunities.
- Scaled the platform by sourcing 30+ alumni interviews (2,800+ Medium views) and spearheading forums for 300+ attendees, growing the community to 480+ followers and generating 40,000+ monthly impressions.

AWARDS & SCHOLARSHIPS

Academia Sinica Scholarship (2025): Awarded funding to support robot research.

Academic Achievement Award, NTU (Fall 2024, Spring 2025): Awarded to top 5% of students each semester.

Academic Excellence Award, NTHU (Fall 2021, Spring 2022): Awarded to top 5% of students each semester.

SKILLS

Programming: Python, Arduino

Robotics & Simulation: Unity, ROS2, CARLA

Computer Vision: OpenCV, MediaPipe, PyTorch

Design: SolidWorks, Autodesk Inventor, AutoCAD, 3D Printing (FDM, SLA)

Language: English (Fluent; TOEFL iBT 100), Mandarin (Native)